

HridAI · system architecture

Heart-failure digital twin for the 30 days after hospital discharge · Team Octas, XLRI Jamshedpur · Happiest Health Digital Twin Challenge 2026

 Built in this proof-of-concept  Production roadmap

1 · DATA STREAMS

Static: EHR / discharge record
Age, sex, ejection fraction, eGFR, NT-proBNP, sodium, diabetes, hypertension, furosemide dose, prior HF admissions, polygenic risk percentile (genetic marker). *Synthetic, generated in `twin.py`.*

Dynamic: smartwatch (nightly)
Night heart rate, HRV (RMSSD), breathing rate, SpO2, sleep efficiency, steps. 10% missing days.

Dynamic: smart scale (daily)
Morning weight, with scale/clothing offsets and 25% skipped days.

Roadmap sources
Synthea FHIR bundles, MIMIC-IV, ABHA-linked records via ABDM; Apple Health / Google Fit / Health Connect; 3-tap WhatsApp symptom check.

2 · INGEST

Clean & align
One row per patient-day; missing values kept as gaps, never imputed.

EHR → twin priors
EF sets heart-rate response to fluid; EHR risk score sets how quickly the twin believes a rising trend.

LLM extraction
Free-text discharge summary → FHIR (Patient, Condition, Observation, MedicationStatement).

Consent & privacy
ABDM consent manager; DPDP Act-compliant storage; de-identified analytics.

3 · PATIENT DIGITAL TWIN (PER PATIENT)

1 Sync · days 0-2 at home learn the person's own baseline for every sensor.

2 State estimation (Kalman filter) · hidden state = [excess fluid L, trend L/day] with uncertainty. Every sensor is a noisy view of that one state.
$$x_{t+1} = F x_t + w \quad \cdot \quad z_t = \text{baseline} + H(\text{EHR}) x_t + v$$

3 5-day forecast · $P(\text{fluid} \geq 2 \text{ L within 5 days})$, closed form from the projected mean and variance. Alert at $\geq 60\%$.

4 Twin fidelity · each day the twin predicts the next readings and scores how often reality lands inside its own 95% band.

5 Active sensing · uncertain + rising + scale skipped → ask the family for a weigh-in.

6 What-if simulation · project fluid for 10 days under a clinician-chosen diuretic change.

4 · PEOPLE WHO ACT

Clinician dashboard (Streamlit)
Belief band, risk, fidelity, alerts vs the 2 kg rule, EHR panel, day-by-day replay, what-if panel.

Caregiver message
Plain language, Hindi + English (templates in PoC).

Delivery
WhatsApp / SMS / IVR in Indian languages; nurse call-list for heart-failure clinics; EHR write-back as FHIR Observation.

Safety
Decision support only. The twin never changes treatment; the clinician decides.

Evaluation pipeline (`make_figures.py`) · dev cohort (seed 7) picks the alert threshold; untouched test cohort (seed 11, 500 patients) reports sensitivity 2-14 days ahead, lead time, false alerts per patient-month vs the 2 kg / 3 day guideline rule; ablations by data stream.

Synthetic cohort realism · person-specific sensor responses, fast and slow decompensators, salty-meal fluid bumps, scale offsets, missing days, and other-cause readmissions the twin should not claim.

Stack · Python, NumPy, pandas, Plotly, Streamlit, Matplotlib, pytest (10 tests incl. no look-ahead leakage). MIT licence. Runs on a laptop in under a minute.